Session 3: The Architecture of Species Distribution Models

Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Model fitting

Correlative Algorithms: From GLMs to Machine Learning

The Core Objective: Construct a mathematical function that estimates how variables (e.g., temperature, salinity, depth) constrain or promote a species' distribution.

The Machine Learning Revolution: Generalized Linear Models (GLMs) - highly interpretable and statistically grounded - can struggle to capture complex, non-linear ecological niches. Advanced Machine Learning techniques can learn complex data.

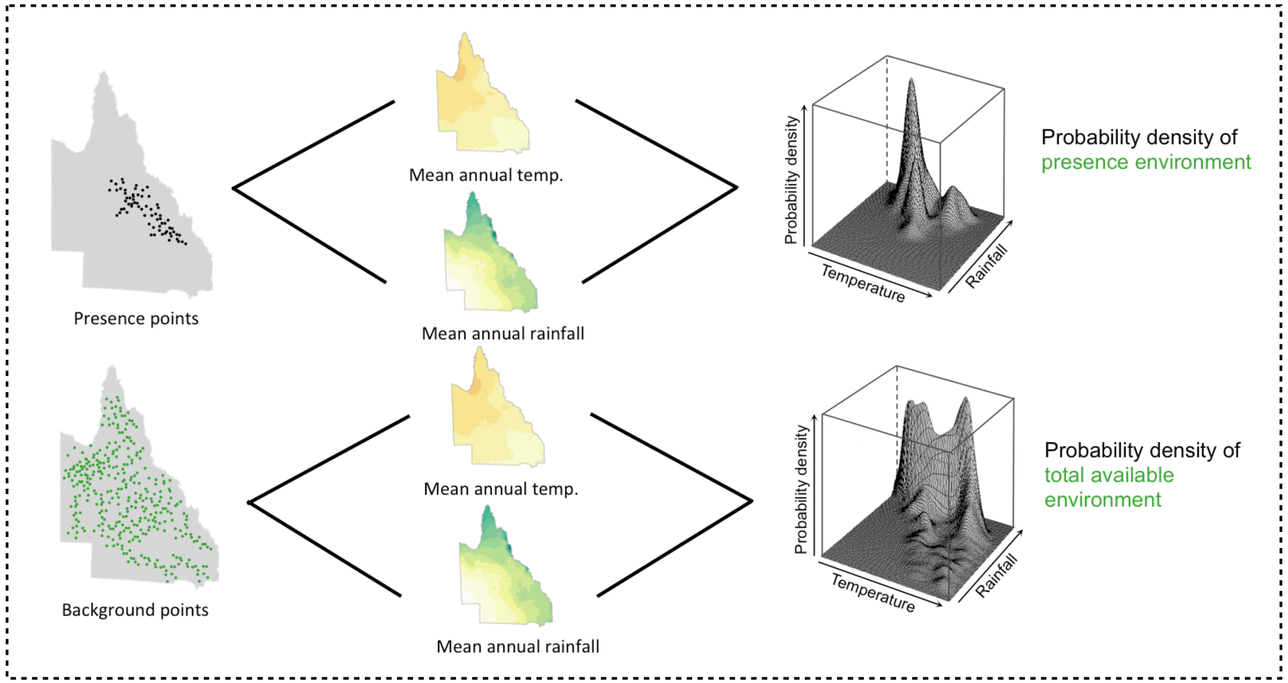
The Cutting Edge: We are currently witnessing an expansion into **Deep Learning** (Neural Networks) to handle massive spatial datasets. Furthermore, we are pushing beyond simple binary predictions (Presence/Absence) to spatially model continuous biological metrics, such as **species Abundance and Biomass**.

The Power of Machine Learning

Exceptional Predictive Performance: Algorithms such as Maximum Entropy (MaxEnt), Boosted Regression Trees (BRT), and Extreme Gradient Boosting (XGBoost) consistently **yield extremely high predictive accuracy**.

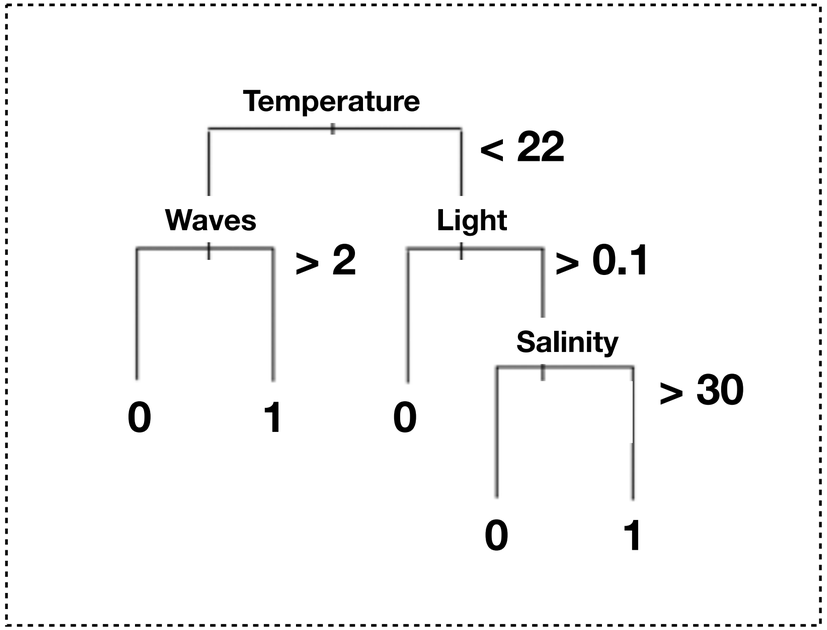
Can seamlessly **integrate both continuous** variables (e.g., temperature gradients) and **categorical variables** (e.g., substrate type).

Can automatically **detect complex variable interactions** - e.g., a species' tolerance to low oxygen severely decreases as water temperature increases (due to higher metabolic demand) - without the researcher having to manually flag interactions.



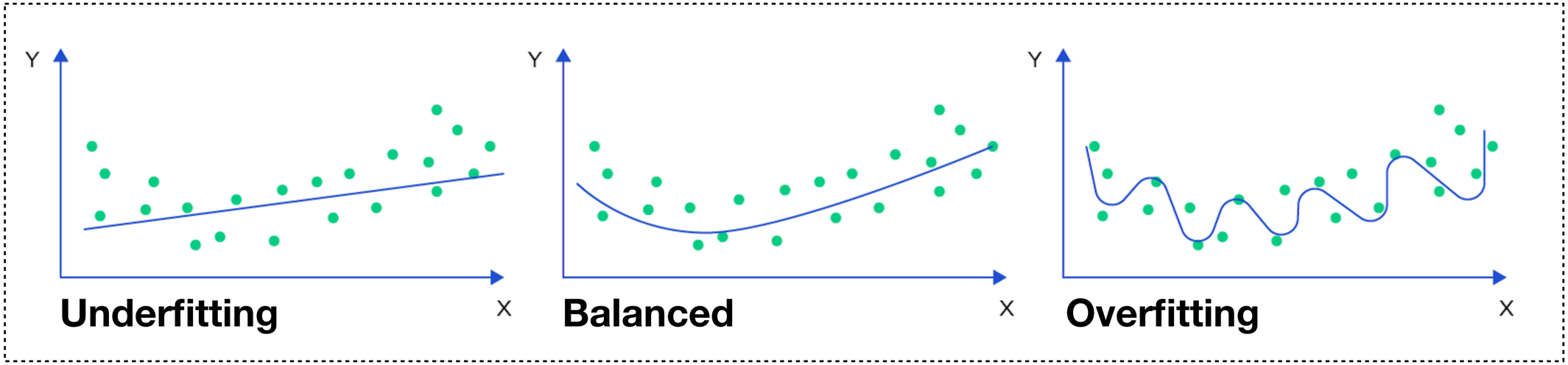
Maximum Entropy fitting made by **comparing the density in the environmental conditions at the locations where the species has been found, with the environmental conditions across the study region**.

MaxEnt **outputs the environmental suitability of a species in the study area** (i.e., the potential distribution).



Boosted Regression Trees is a flexible and high performance algorithm, which combine the strengths of **decision trees** (models that relate a response to their predictors by recursive binary splits) **with boosting** (combines simple models to give improved predictive performance).

BRT gives the probability of a species occurring in the study area.

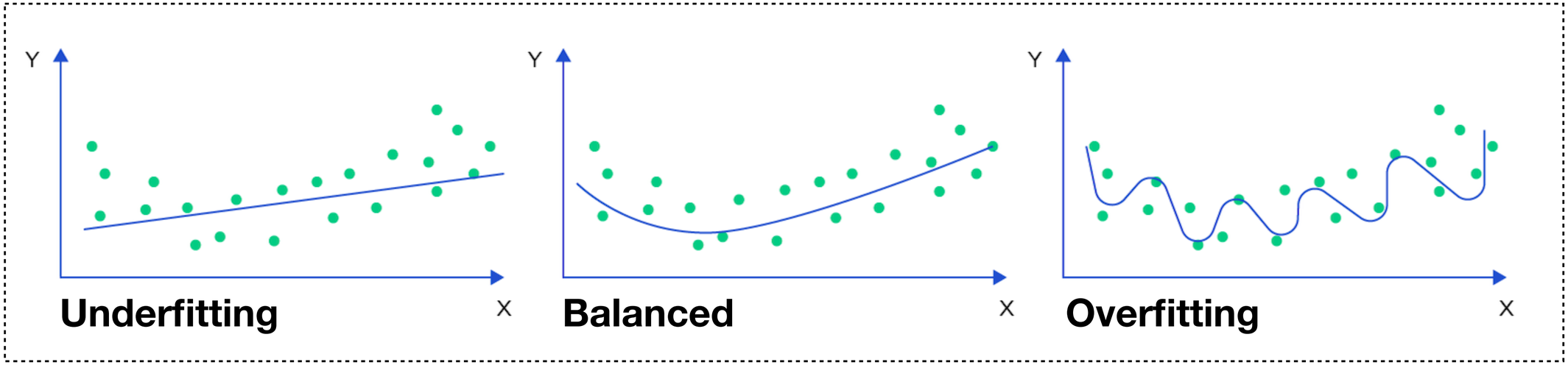


Model fitting pitfalls: underfitting

The model **is too simple to capture the underlying patterns in the data**.

Sources: simplistic algorithms (e.g., a GLM for complex relationships). Few or irrelevant variables. Insufficient occurrence data to represent the species' niche. Excessive regularization during model fitting.

Issues: Poor predictive performance. Species-environment relationships are missed, leading to unrealistic outputs.



Model pitfalls: overfitting

The **model learns the training data too well**, **including random noise and very specific patterns of the sample**, rather than the general patterns.

Sources: overly flexible model algorithms. Including too many environmental variables. Not applying appropriate regularization during model training.

Issues: performs poorly on new data (e.g., future). Creates complex, biologically implausible distribution maps (e.g., distributions in tiny, isolated pockets based on noise).

Desired properties of model fitting

Deductive and general: to corroborate hypotheses around the causes of the pattern explained by the model, aiming at achieving ecological conclusions.

Simple: highlight a few mechanisms without becoming entangled in complex interactions between variables.

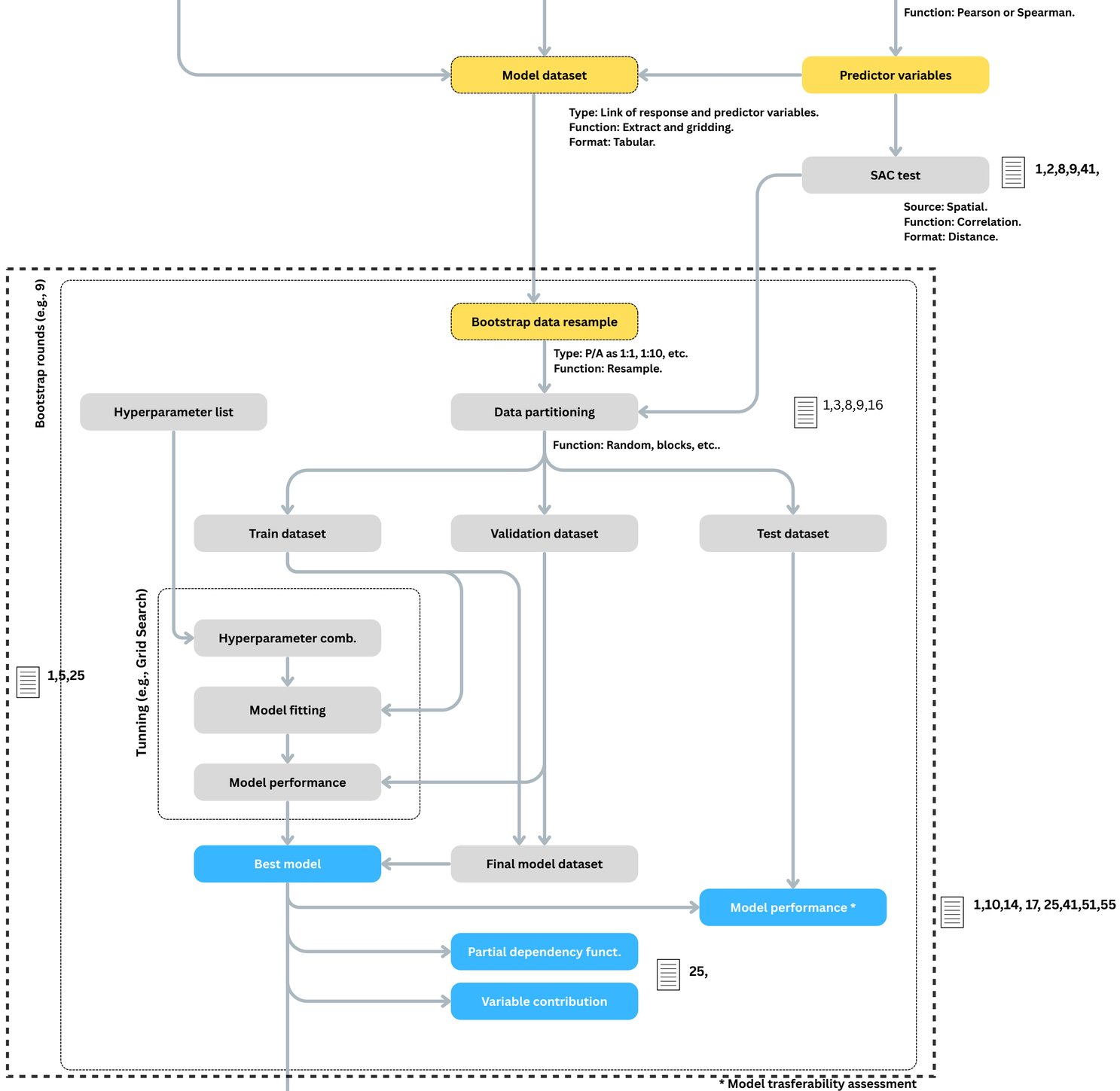
Parsimonious: preferring the simpler of two equally adequate models; favor simple explanations over complex ones.

The Danger of Algorithm Default Settings

The "Black Box" Trap: Because ML algorithms are so powerful, they are highly susceptible to overfitting if left completely unrestricted.

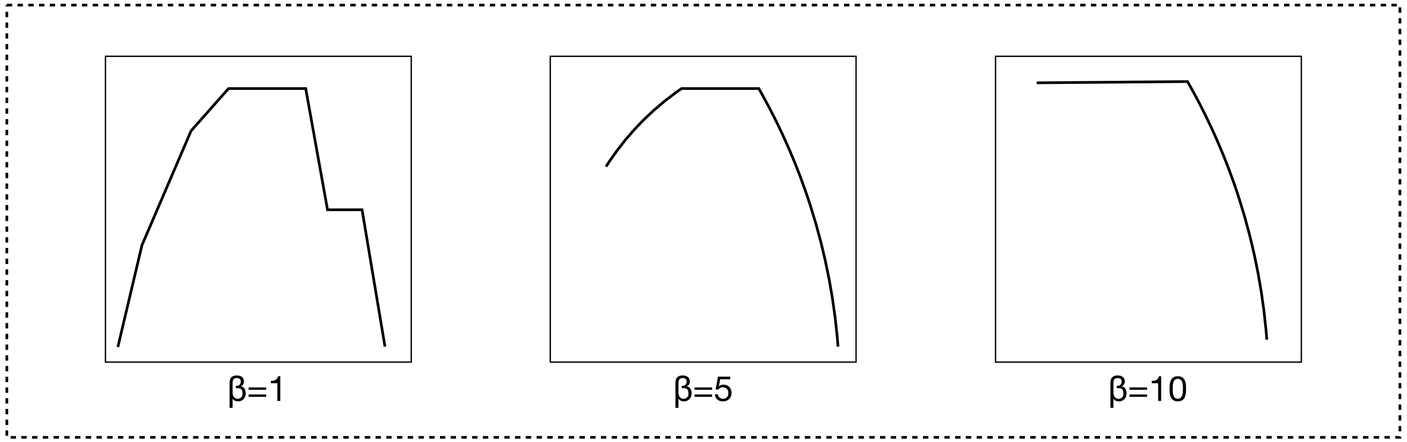
Relying on the"default settings”, i.e., "default Hyperparameters" is a massive pitfall. There is no universal "rule of thumb" because every dataset is unique.

The Solution: Hyperparameter Tuning - researchers must actively constrain the algorithms by tuning their hyperparameters.


Function: Pearson or Spearman.
Model dataset
Predictor variables
Type: Link of response and predictor variables.
Function: Extract and gridding.
Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
Bootstrap data resample
Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..
Train dataset
Validation dataset
Test dataset
Tunning (e.g., Grid Search)
Hyperparameter comb.
1,5,25
Model fitting
Model performance
Best model
Final model dataset
Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
25,
Variable contribution
* Model trasferability assessment

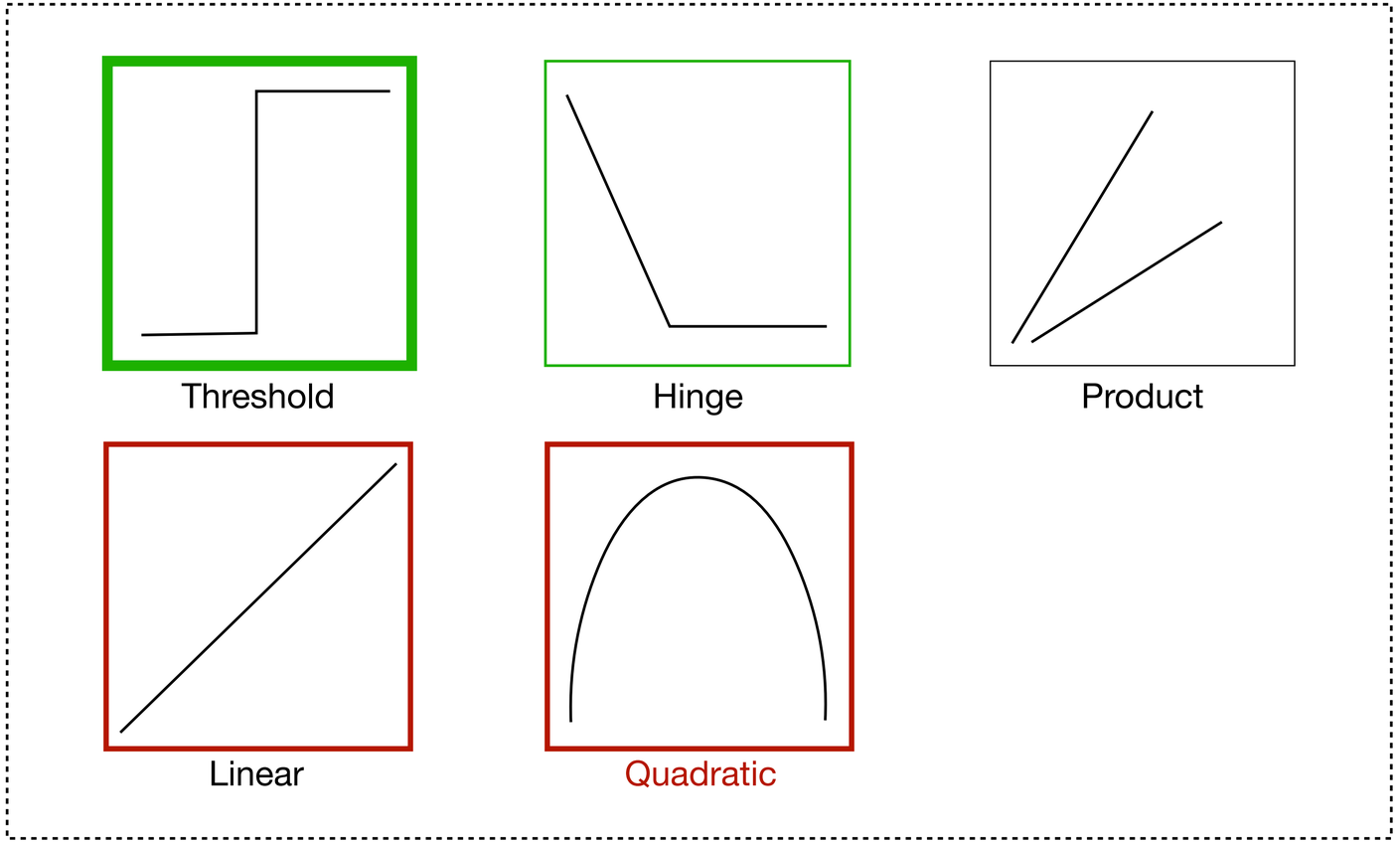
MaxEnt hyperparameters

A key parameter of **MaxEnt** is **regularization**, which controls **fit constraints and therefore strongly impacts overfitting**.



Higher regularization (β multiplier) values reduced the smoothing / relaxation of curves, which in term, can reduce overfitting.

To much regularization can lead to underfitting.

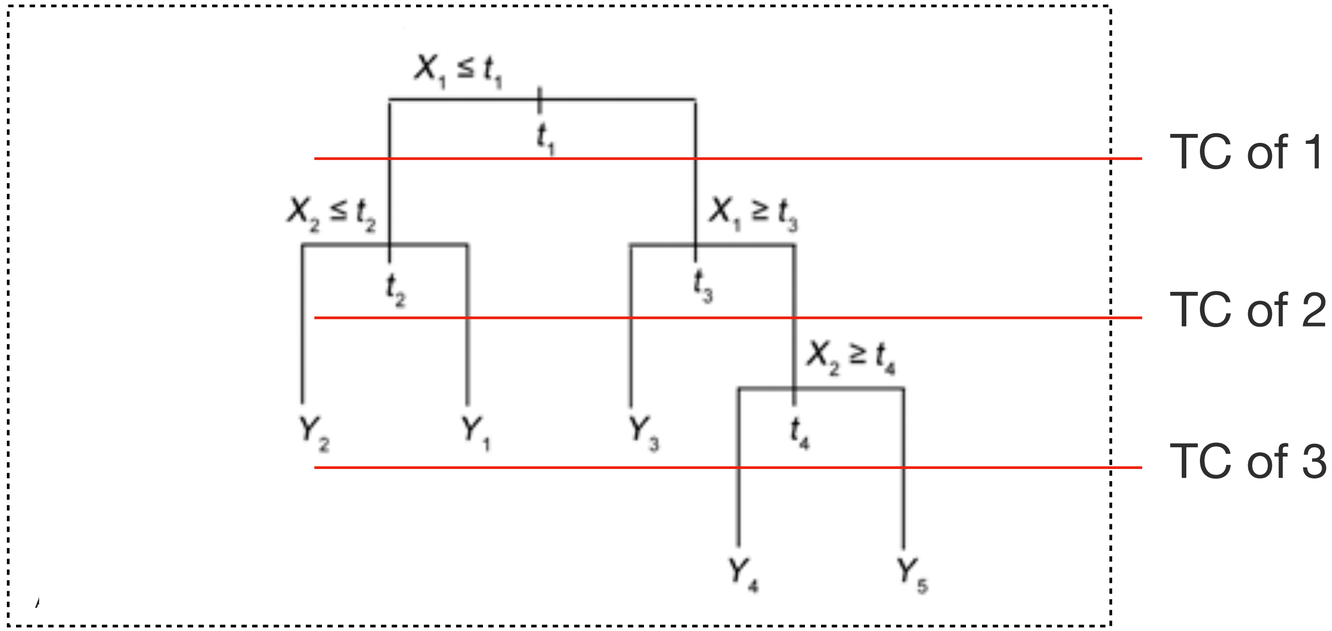


A key parameter of MaxEnt is **feature type - the type of the function**.

We tend to **exclude features that are more prone to lead to overfitting or to underfitting.**

Boosted Regression Trees hyperparameters

Tree complexity: a TC of 1 (single decision; two terminal nodes), a TC of 2 fits a model with up to two-way interactions, and so on. **TC controls complexity**, higher / lower can translate into overfitting / underfitting.

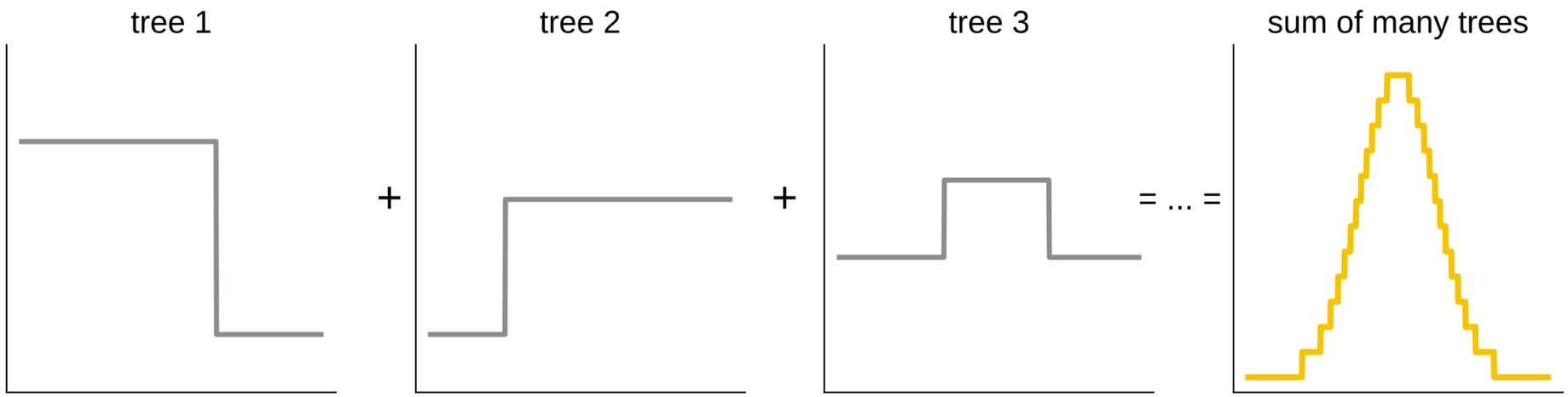


Boosted Regression Trees hyperparameters

Learning rate: the contribution of each tree to the final model;

Number of trees: the number of individual trees included in the final model;

A smaller lr with more trees is generally preferable.

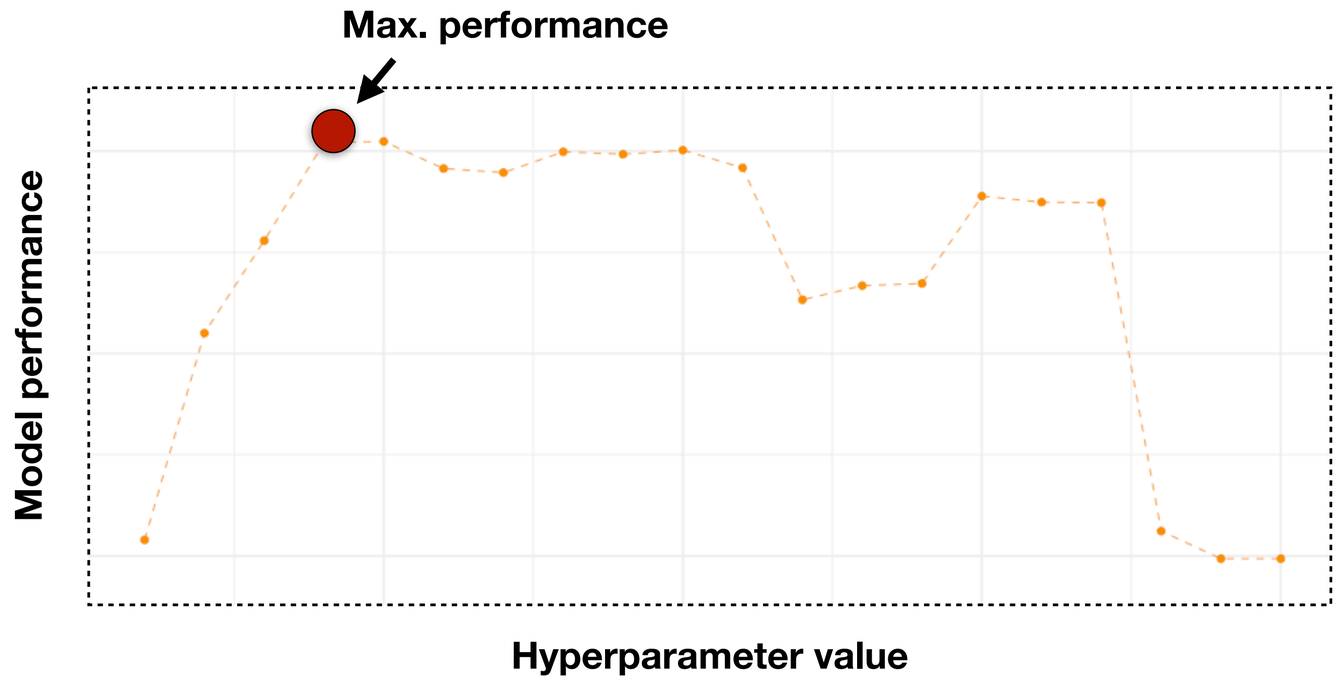


Hyperparameter optimization

How to tune these parameters to improve model fitting?

The approach relies on testing a span of parameter values and finding which value maximizes the performance of models.

Which parameter value leads to higher model performance?



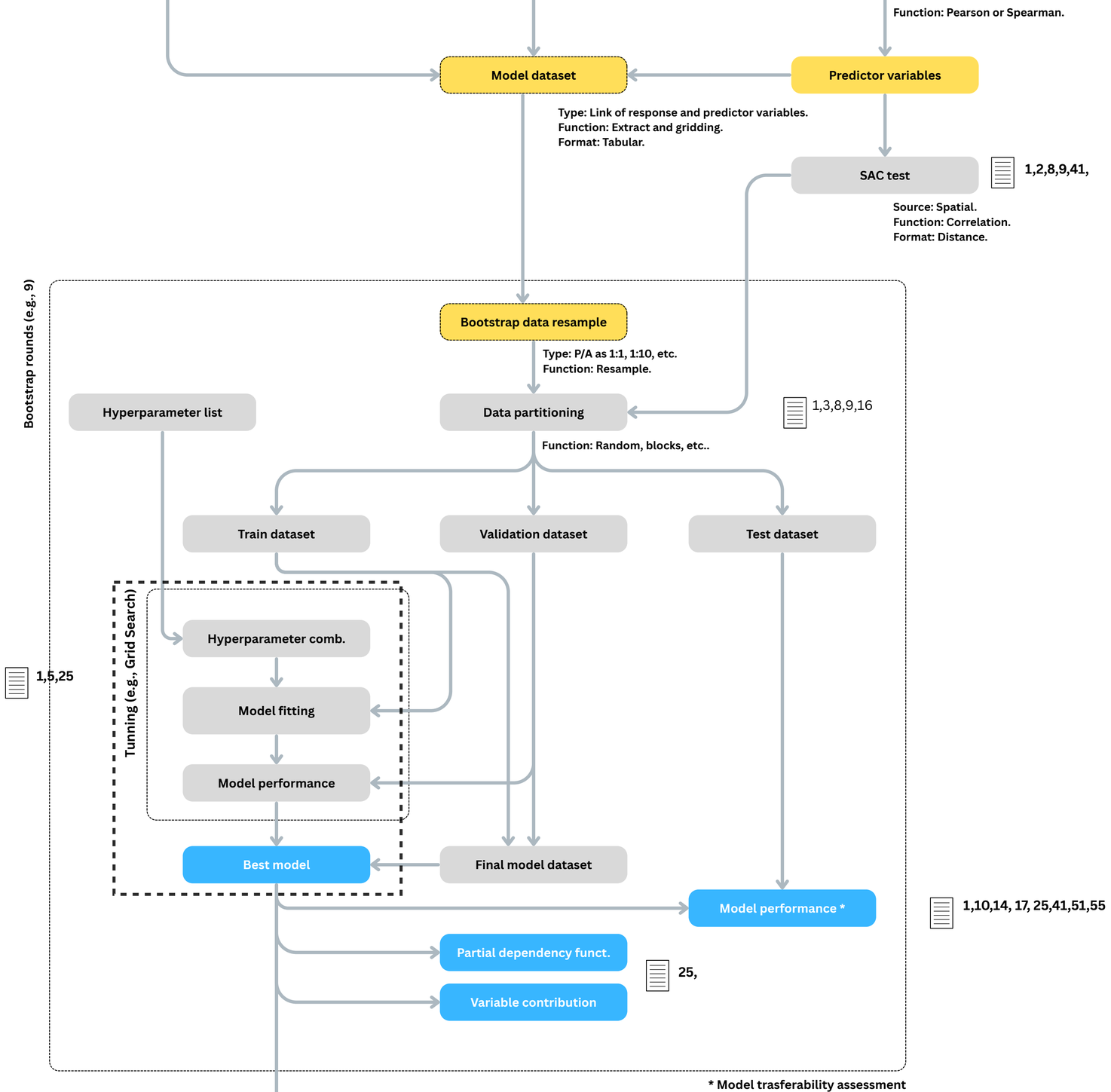
Hyperparameter optimization

How to tune these parameters to improve model fitting?

Grid search: Grid: every combination of candidate values,

e.g., *3 learning rates × 3 depths × 10 numbers of trees = 90 models per round. Keep the combination with the highest validation performance.*

Comb	TC	LR	N	Performance
1	2	0.01	100	0.7
2	3	0.001	200	0.8
3	4	0.0001	300	0.7
4	5	0.00001	400	0.6


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